# MARKET REGIME DETECTION SYSTEM

A Complete Plain-English Guide

How the system identifies Bull · Sideways · Bear market states

K-Means Clustering · XGBoost Classifier · LSTM Autoencoder
Mathematics · Features · Reliability · Manual Quant Extensions

## Contents

## 1. What is a Market Regime?

A **market regime** is the dominant behavioural state of a financial market over a sustained period. Just as weather has seasons — summer, monsoon, winter — financial markets move through recurring structural states that differ in their average returns, volatility levels, and directional momentum.

Identifying the current regime matters because **different regimes demand different portfolio strategies**. A position that performs well in a strong bull market may be catastrophic during a bear phase. This system identifies three regimes:

| BULL REGIME | SIDEWAYS REGIME | BEAR REGIME |
|---|---|---|
| **Rising market.** Prices trend upward consistently, investor confidence is high, and volatility is relatively low. Momentum is positive across multiple timeframes.<br><br>*Example: Nifty 500 rally of 2021, when the index rose from 9,000 to 18,000.* | **Range-bound market.** Prices oscillate within a horizontal channel. No clear directional bias. Investors are uncertain and waiting for a clear catalyst.<br><br>*Example: Mid-2023, when Nifty traded between 17,000 and 19,500 for several months.* | **Falling market.** Prices trend downward, selling pressure dominates, volatility spikes, and negative skewness increases as crash risk rises.<br><br>*Example: March 2020 COVID sell-off — Nifty fell approximately 40% in six weeks.* |

**Note:** Looking at a single day's return is insufficient. A +2% return can occur within a bear regime — it may simply be a short-covering bounce, not a genuine reversal. This system uses multiple signals across return, volatility, momentum, and machine learning probability to make a robust regime determination.

## 2. System Architecture — Two-Stage Overview

The regime detection engine operates in two sequential stages. Each stage has a distinct role; together they produce a reliable, probability-backed regime signal.

| STAGE 1 | STAGE 2 |
|---|---|
| **K-Means Clustering** | **XGBoost Classifier** |
| **Type:** Unsupervised learning<br><br>Groups historical trading days into three clusters based on similarity of market features — without any prior labels. The algorithm discovers the natural structure in the data itself.<br><br>**Output:** Each past trading day is assigned a raw cluster label (0, 1, or 2). These are then sorted by average return to map to Bear, Sideways, and Bull. | **Type:** Supervised learning<br><br>Trained on the cluster labels produced by Stage 1 — learning which combinations of 19 market features predict each regime. Unlike K-Means, it outputs a *probability*, not just a label.<br><br>**Output:** P(Bull), P(Sideways), P(Bear) for any given day. The highest probability wins. The value itself is the confidence score. |
| *Analogy: Sort flowers by colour with no prior labels* | *Analogy: Learn from labelled examples to classify new ones* |

**Why both stages?** K-Means provides the historical ground truth labels — it discovers the natural clusters in the data without any assumptions. XGBoost then learns from those labels to produce probability estimates on new data. Neither model alone is sufficient: K-Means cannot output probabilities; XGBoost needs labelled training data. Together, the combination is robust.

# 3. Stage 1: K-Means Clustering — Grouping Market Days

K-Means is an **unsupervised** algorithm. It is given a set of data points (trading days, each described by 7 numbers) and asked to divide them into 3 groups such that days within each group are as similar as possible to one another, and as different as possible from other groups.

Think of it as sorting a box of mixed fruit by size and colour — without anyone telling you how many categories to use. The algorithm discovers the natural groupings on its own.

## The 7 Clustering Features

| Feature | Full Name | What It Captures | Typical Range |
|---|---|---|---|
| `ret_21d` | 21-day compound return | Last month's price momentum | –30% to +30% |
| `ret_63d` | 63-day compound return | 3-month trend direction | –40% to +40% |
| `vol_21d` | 21-day annualised volatility | Recent price fluctuation (risk) | 8% to 60% |
| `vol_63d` | 63-day annualised volatility | Medium-term baseline risk | 8% to 50% |
| `vol_ratio` | vol_21d ÷ vol_63d | Rising/falling volatility regime | 0.4 to 2.5 |
| `skew_21d` | 21-day return skewness | Crash risk (negative = left tail) | –3 to +3 |
| `ret_z21` | Z-score of 21-day return | Momentum relative to 6-month avg | –3 to +3 |

## The K-Means Algorithm — Step by Step

| Step | Action | Output |
|---|---|---|
| 1 | Normalise all 7 features using StandardScaler (zero mean, unit variance) | **Scaled feature matrix** |
| 2 | Randomly initialise 3 centroids. Repeat 20 times with different seeds (n_init=20) to avoid local minima | **3 starting centroids** |
| 3 | Assign each trading day to its nearest centroid using Euclidean distance | **Initial cluster labels** |
| 4 | Recompute each centroid as the mean of all days in that cluster | **Updated centroids** |
| 5 | Repeat steps 3–4 until cluster assignments no longer change (max 500 iterations) | **Stable clusters** |
| 6 | Sort clusters by average ret_21d — lowest average → Bear (0), middle → Sideways (1), highest → Bull (2) | **Named regime labels** |

**Note:** Setting n_init=20 means the algorithm runs 20 times from different random starting positions and selects the result with the lowest total intra-cluster variance. This dramatically reduces the risk of converging on a poor local solution.

# 4. The Mathematics of K-Means

Each formula below is first explained in plain English, then written formally. No prior mathematical background is required to understand the logic.

## Formula 1 — Euclidean Distance

**Plain English:** To assign a trading day to a cluster, we measure how far it is from each of the 3 cluster centres. Distance is computed across all 7 feature dimensions simultaneously — like measuring distance in 7-dimensional space rather than on a flat map.

```
dist(day_i, centroid_k) = sqrt[ (f1_i - f1_k)^2 + (f2_i - f2_k)^2 + ... + (f7_i - f7_k)^2 ]
```

f1...f7 = the 7 features | i = a trading day | k = cluster index (0, 1, or 2)

**Example:** Today ret_21d = +14%, vol_21d = 12%. Bull centroid: ret_21d = +11%, vol_21d = 11%. Bear centroid: ret_21d = –9%, vol_21d = 31%. Today is much closer to the Bull centroid → assigned to Bull cluster.

## Formula 2 — Objective Function (What K-Means Minimises)

**Plain English:** K-Means tries to build groups where all the members within each group are as tightly clustered together as possible. The tighter the groups, the more distinct the regimes.

```
Minimise: sum over k [ sum over all days_i in cluster_k [ dist(day_i, centroid_k)^2 ] ]
```

This is called 'intra-cluster variance'. Smaller = tighter, more distinct clusters.

## Formula 3 — Centroid Update Rule

**Plain English:** After assigning days to clusters, each cluster's centre is recalculated as the simple average of all its member days across all 7 features. This is repeated until the assignments stabilise.

```
new_centroid_k = (1 / |C_k|) * sum of all feature vectors in cluster C_k
```

|C_k| = number of days currently in cluster k

## Formula 4 — StandardScaler Normalisation

**Why this is essential:** K-Means uses Euclidean distance, which is sensitive to the scale of each feature. If vol_21d ranges 0.08–0.50 while ret_21d ranges –0.30 to +0.30, the distance calculation would be dominated by whichever feature has the larger numerical range. Normalising to the same scale gives every feature equal weight.

```
normalised_value = (original_value - feature_mean) / feature_std_dev
```

Result: every feature has mean = 0, standard deviation = 1

| Feature | Raw Value | Feature Mean | Feature Std | Normalised Value |
|---|---|---|---|---|
| ret_21d | +14.2% | 2.1% | 9.3% | (0.142 - 0.021) / 0.093 = +1.30 |
| vol_21d | 13.1% | 18.4% | 7.8% | (0.131 - 0.184) / 0.078 = –0.68 |
| skew_21d | +0.42 | –0.18 | 0.71 | (0.42 + 0.18) / 0.71 = +0.85 |
| vol_ratio | 0.78 | 1.02 | 0.31 | (0.78 - 1.02) / 0.31 = –0.77 |

# 5. Stage 2: XGBoost Classifier — Learning Probabilities

XGBoost (*Extreme Gradient Boosting*) is an ensemble model consisting of hundreds of small decision trees built sequentially. Each tree corrects the errors of the previous ones. The result is a powerful classifier that produces calibrated probability estimates for each regime.

## Conceptual Analogy

Imagine a panel of 400 junior analysts. Each analyst studies one narrow aspect of the market — Analyst 1 looks only at 3-month momentum, Analyst 2 at volatility ratio, Analyst 3 at RSI, and so on. Each gives a partial opinion. XGBoost aggregates all 400 opinions into a final probability-weighted verdict. No single analyst is decisive; the panel as a whole is highly accurate.

## How XGBoost is Trained

Stage 1 (K-Means) provides labelled training data: each historical day is tagged as Bear, Sideways, or Bull. XGBoost is given the full set of 19 features for each day and learns which feature patterns correspond to which regime label.

| Iteration | What the Tree Learns | Accuracy Gain |
|---|---|---|
| `Tree 1` | If ret_63d > 10% → likely Bull | **Base prediction** |
| `Tree 2` | Residuals from Tree 1: vol_21d also matters | **+4–5%** |
| `Tree 3` | Skewness adds crash-risk signal for Bear | **+2–3%** |
| `Tree 4–50` | Refine interactions between features | **Cumulative** |
| `Tree 51–400` | Fine-tune edge cases and low-confidence days | **Best overall** |

## Model Configuration and Why

| Parameter | Value | Rationale |
|---|---|---|
| `n_estimators` | `400 trees` | Sufficient capacity to model non-linear interactions |
| `max_depth` | `4 levels` | Prevents overfitting to training data |
| `learning_rate (eta)` | `0.03` | Small steps = better generalisation (shrinkage) |
| `subsample` | `0.80` | Each tree sees 80% of rows = reduces variance |
| `colsample_bytree` | `0.80` | Each tree sees 80% of features = decorrelated trees |
| `min_child_weight` | `5` | Minimum 5 observations per leaf = stability |
| `eval_metric` | `log-loss` | Optimises probability calibration, not just class label |
| `CV strategy` | `TimeSeriesSplit (5-fold)` | Walk-forward: no future data leaks into training |

## XGBoost Output — Probability Vector

Unlike K-Means (which returns a single label), XGBoost returns a **probability vector** for every day:

| Regime | Example Probability | Interpretation |
|---|---|---|
| Bull | P = 0.76 (76%) | 76% model confidence that today is a Bull regime |
| Sideways | P = 0.18 (18%) | 18% probability of Sideways |
| Bear | P = 0.06 (6%) | 6% probability of Bear regime |
| **→ RESULT** | **BULL — 76% confident** | **Highest probability wins. Confidence = 76%.** |

**Note:** Confidence threshold rule: If max(P) < 0.65, the model is uncertain. The system defaults to Sideways (the conservative choice) regardless of which class has the highest individual probability. Example: P(Bull)=0.48, P(Side)=0.38, P(Bear)=0.14 → classified as Sideways.

# 6. How Bull / Bear / Sideways is Decided

Regime classification uses a **two-layer confirmation** approach. The ML signal (XGBoost probability) is the primary input. Technical indicators then act as a cross-check layer. When both agree, confidence is high.

## Bull Regime — Confirmation Checklist

| Signal | Required Condition | Why This Matters |
|---|---|---|
| `ret_21d` | > +5% | Positive monthly momentum — uptrend active |
| `ret_63d` | > +8% | 3-month trend confirms sustained move, not noise |
| `vol_21d` | < vol_63d | Recent vol lower than baseline — market calm |
| `vol_ratio` | < 0.90 | Short-term risk falling → complacency = bull fuel |
| `skew_21d` | > −0.5 | No significant crash risk visible in return tail |
| `EMA 20 vs 60` | EMA20 > EMA60 | Golden Cross: short trend above long trend |
| `MACD Histogram` | > 0 | Price momentum is net positive |
| `RSI-14` | 50 – 70 | Strong but not overbought — healthy bull |
| `XGBoost P(Bull)` | ≥ 65% | Primary ML model is confident |

## Bear Regime — Confirmation Checklist

| Signal | Required Condition | Why This Matters |
|---|---|---|
| `ret_21d` | < −5% | Negative monthly momentum — downtrend active |
| `ret_63d` | < −8% | 3-month trend confirms sustained deterioration |
| `vol_21d` | > vol_63d × 1.3 | Volatility spiking relative to baseline — panic |
| `skew_21d` | < −1.0 | Heavy left tail — crash risk visible in distribution |
| `EMA 20 vs 60` | EMA20 < EMA60 | Death Cross: short trend below long trend |
| `MACD Histogram` | < 0 | Price momentum net negative |
| `RSI-14` | < 40 | Weak market — approaching oversold territory |
| `XGBoost P(Bear)` | ≥ 65% | Primary ML model is confident of Bear regime |

## Sideways Regime — When It is Declared

Sideways is declared in three situations: (1) neither Bull nor Bear conditions are met, (2) XGBoost confidence is below 65%, or (3) the manual scorecard (Section 9) falls between –5 and +5. It is the system's way of saying: **the evidence is mixed — take a cautious stance.**

Sideways indicators: |ret_21d| < 5% · ADX < 20 (weak trend) · vol_ratio between 0.90 – 1.10 · RSI between 45 – 55 · MACD Histogram near zero

## 7. All 19 Features — What Each One Measures

XGBoost is trained on 19 features covering four distinct dimensions of market behaviour: momentum, volatility, statistical distribution, and technical indicators. No single feature tells the full story; the model learns which combinations matter most through training.

| Category | Feature | Formula | What It Detects |
|---|---|---|---|
| **Momentum** | **ret_5d** | Compound 5-day return | 1-week price direction |
| **Momentum** | **ret_10d** | Compound 10-day return | 2-week price direction |
| **Momentum** | **ret_21d** | Compound 21-day return | Monthly trend — most predictive |
| **Momentum** | **ret_42d** | Compound 42-day return | 6-week intermediate trend |
| **Momentum** | **ret_63d** | Compound 63-day return | Quarterly trend — #1 by SHAP |
| **Momentum** | **ret_126d** | Compound 126-day return | 6-month long-run momentum |
| **Volatility** | **vol_10d** | Std dev (10 days) × sqrt(252) | Short-term risk spike detector |
| **Volatility** | **vol_21d** | Std dev (21 days) × sqrt(252) | Monthly volatility — crash flag |
| **Volatility** | **vol_63d** | Std dev (63 days) × sqrt(252) | Medium-term baseline risk |
| **Volatility** | **vol_ratio** | vol_10d ÷ vol_63d | Volatility regime shift signal |
| **Distribution** | **skew_21d** | E[(r−mu)^3] / sigma^3 over 21 days | Negative = left tail = crash risk |
| **Distribution** | **kurt_21d** | E[(r−mu)^4] / sigma^4 over 21 days | Fat tails = extreme moves likely |
| **Technical** | **rsi_14** | 100 − 100/(1 + avg_gain/avg_loss) over 14 days | Overbought (>70) / Oversold (<30) |
| **Technical** | **rsi_21** | Same formula over 21 days | Slower, less noise-prone RSI |
| **Technical** | **macd_h** | MACD Line − Signal Line (12, 26, 9 params) | Momentum acceleration |
| **Technical** | **bb_pct** | (Close − Lower BB) / (Upper BB − Lower BB) | Price position within BB channel |
| **Technical** | **ema_gap** | (EMA20 − EMA60) / EMA60 | Golden / Death Cross magnitude |
| **Technical** | **adx** | Average Directional Index (14-period) | Trend strength irrespective of direction |
| **Mean Reversion** | **ret_z21** | (ret_21d − rolling_mean_126) / rolling_std_126 | Short-term Z-score — mean reversion signal |

**Note:** SHAP analysis consistently ranks ret_63d (3-month momentum) as the most important feature. This is consistent with a large body of academic literature — the momentum factor is one of the most robust return predictors in equity markets globally.

# 8. System Reliability — An Honest Assessment

**Important: No model achieves 100% accuracy in financial markets. 60–65% directional accuracy is considered excellent in quantitative finance. This system targets 58–65% on 21-day direction prediction using walk-forward cross-validation (TimeSeriesSplit).**

## What is Reliable — What is Not

| Component | Reliability | Explanation |
|---|---|---|
| **Bull vs Bear separation** | High (~72%+) | Extreme states are structurally distinct — easy to separate |
| **Regime direction (Bull/Bear)** | Medium–High | Strong trending markets are captured accurately |
| **Regime transition timing** | Low–Medium | Exact turning points are hard to predict in any model |
| **Confidence probability score** | High | XGBoost with log-loss objective produces calibrated probabilities |
| **LSTM anomaly detection** | High (extremes) | 2008-style and 2020-style crashes are reliably flagged |
| **Sideways regime detection** | Medium (~60%) | Sideways is the most ambiguous state by definition |
| **Short-term noise (1–3 day moves)** | Low | Not designed for high-frequency or intraday signals |

## Five Techniques to Improve Reliability

**1. 5-Day Persistence Filter**
A regime change must persist for at least 5 consecutive trading days before it is accepted as a genuine shift. One or two days of contradictory signals are treated as noise. Real regime transitions take time to establish.

**2. 65% Confidence Threshold**
If the highest XGBoost probability is below 65%, the system defaults to Sideways. Acting on an uncertain signal is riskier than waiting. Missing some gains in a weak bull signal is preferable to taking full exposure on an ambiguous one.

**3. Multi-Timeframe Confirmation**
Cross-check the daily regime signal against a weekly timeframe calculation. If daily says Bull but weekly says Sideways, reduce conviction and deployed capital proportionally. The weekly signal filters out short-term noise.

**4. Silhouette Score Monitoring (K-Means Quality)**
Silhouette Score measures how well-separated the K-Means clusters are. Score > 0.5 = well-separated clusters (reliable). Score < 0.3 = overlapping clusters (unreliable). Re-run K-Means on fresh data monthly or when the score drops.

**5. Minimum Training Data Requirement**
K-Means requires at least 500 trading days (~2 years) to form stable, meaningful clusters. This system uses 5 years of data, which is adequate. Using less data risks clusters that overfit to a specific recent period.

## Cross-Validation Methodology

The direction prediction model is validated using **TimeSeriesSplit** with 5 folds. Unlike standard k-fold cross-validation (which would allow future data to leak into training), TimeSeriesSplit respects the temporal order of the data:

| Fold | Training Window | Validation Window | Purpose |
|---|---|---|---|
| 1 | Days 1 – 400 | Days 401 – 450 | Early period test |
| 2 | Days 1 – 550 | Days 551 – 600 | Expanding window |
| 3 | Days 1 – 700 | Days 701 – 750 | Mid period test |

| | | | |
|---|---|---|---|
| 4 | Days 1 – 850 | Days 851 – 900 | Recent test |
| 5 | Days 1 – 1000 | Days 1001 – 1050 | Final out-of-sample |
| **Avg** | **—** | **Mean accuracy across all 5** | **Reported CV accuracy** |

# 9. Manual Quantitative Extensions — What Can Be Added

The ML models provide a strong quantitative foundation. However, they operate exclusively on price and return data. The following extensions introduce macro-economic, breadth, sentiment, and structural signals that are **not captured by price alone**. Each adds an independent information source.

## Category A — Macro-Economic Indicators (Leading Signals)

These signals originate outside of price data and often lead market movements by weeks or months. They provide context that pure price models cannot see.

| Indicator | Bull Signal | Bear Signal | Data Source |
|---|---|---|---|
| **India 10Y vs 1Y Bond Yield (Yield Curve)** | Spread widening (> 1%) | Curve inverts (10Y < 1Y) | RBI / CCIL daily |
| **US Fed Funds Rate Decision** | Rate cuts / pause | Rate hikes | FOMC calendar |
| **India PMI (Manufacturing)** | PMI > 50 for 3 months | PMI < 50 for 3 months | S&P Global monthly |
| **FII Net Flow (5-day)** | Net buyer > INR 2,000 Cr | Net seller | NSE daily |
| **India CPI Inflation** | CPI < 4.5% | CPI > 6.5% sustained | MOSPI monthly |
| **USD/INR Exchange Rate** | Rupee stable / gaining | Depreciation > 2%/month | RBI / Bloomberg |

## Category B — Market Breadth Indicators

Breadth measures how *widely* market moves are distributed across individual stocks. A rally in which 80% of stocks are participating is structurally stronger than one driven by 10 large-cap stocks.

| Indicator | Bull Threshold | Bear Threshold | Logic |
|---|---|---|---|
| **Advance-Decline Ratio (daily)** | A/D > 2.0 | A/D < 0.5 | Measures breadth of daily move |
| **% Stocks Above 200-day MA** | > 70% above MA200 | < 40% above MA200 | Long-term participation |
| **New 52-Week Highs vs Lows** | Highs >> Lows | Lows >> Highs | Breadth thrust or distribution |
| **McClellan Oscillator** | Positive and rising | Negative and falling | Breadth momentum |

## Category C — Sentiment Indicators

Sentiment measures the collective emotional state of market participants. Extreme fear often marks buying opportunities; extreme complacency often precedes corrections.

| Indicator | Bull Signal | Bear Signal | Source |
|---|---|---|---|
| **India VIX (Fear Gauge)** | VIX < 15 | VIX > 25 | NSE daily |
| **Put-Call Ratio (PCR)** | PCR 0.70 – 1.00 | PCR > 1.30 | NSE F&O daily |
| **Short Interest Ratio** | Falling short interest | Rising short interest | NSE / Prime DB |
| **Mutual Fund Cash Level** | Low cash deployment | High cash held back | AMFI monthly |

## Category D — Manual Regime Scorecard (Recommended Extension)

This is the most actionable addition. Each indicator is assigned +1 (Bull), –1 (Bear), or 0 (Neutral). The total score maps directly to a regime and a recommended deployment level. It provides a human-readable validation layer over the ML output.

| Indicator | Bull (+1) | Bear (–1) | Neutral (0) |
|---|---|---|---|
| `ret_21d` | > +5% | < –5% | –5% to +5% |
| `vol_ratio` | < 0.85 | > 1.20 | 0.85 – 1.20 |
| `EMA 20 vs 60` | EMA20 > EMA60 | EMA20 < EMA60 | Crossing zone |
| `India VIX` | < 15 | > 25 | 15 – 25 |
| `Put-Call Ratio` | 0.70 – 1.00 | > 1.30 | 1.00 – 1.30 |
| `Advance-Decline` | A/D > 1.8 | A/D < 0.6 | 0.6 – 1.8 |
| `MACD Histogram` | > 0 | < 0 | Near zero |
| `RSI-14` | 55 – 70 | < 40 | 40 – 55 |
| `FII Net Flow 5d` | Net buyer | Net seller | Neutral |
| `XGBoost P (primary)` | P(Bull) ≥ 65% | P(Bear) ≥ 65% | Mixed / < 65% |

**Score Interpretation and Recommended Portfolio Action:**

| | | |
|---|---|---|
| **+8 to +10** | **STRONG BULL** | **Full deployment (100%). Favour high-beta stocks (Beta > 1.2).** |
| **+5 to +7** | **BULL** | **85–100% deployment. Standard EGP allocation.** |
| **+2 to +4** | **MILD BULL** | **75–85% deployed. Moderate position sizing.** |
| **–1 to +1** | **SIDEWAYS** | **65–75% deployed. Wait for clearer directional signal.** |
| **–2 to –4** | **MILD BEAR** | **50–65% deployed. Reduce high-beta exposure.** |
| **–5 to –7** | **BEAR** | **30–50% deployed. Defensive stocks only.** |
| **–8 to –10** | **STRONG BEAR** | **< 30% deployed. Maximum defensive positioning. Preserve capital.** |

**Note:** The scorecard validates the ML output. If XGBoost says Bull but the score is only +3, treat it as a weak bull signal — reduce conviction. High-confidence signals require both ML probability ≥ 65% AND scorecard ≥ +6.

## 10. Complete Decision Flowchart

The following table describes the complete step-by-step process for determining the regime on any given trading day:

| Step | Action | Output |
|---|---|---|
| 1 | Load daily Nifty 500 close prices | **Raw price series** |
| 2 | Compute 7 clustering features using rolling windows | **Feature vectors per day** |
| 3 | Apply StandardScaler normalisation to all 7 features | **Scaled feature matrix** |
| 4 | K-Means: find nearest centroid for today's feature vector | **Raw cluster (0, 1, or 2)** |
| 5 | XGBoost: compute P(Bull), P(Sideways), P(Bear) from 19 features | **Probability vector** |
| 6 | Predicted regime = argmax of probability vector | **Predicted regime label** |
| 7 | Confidence check: max(P) ≥ 65%? NO → default to Sideways | **Accepted or overridden** |
| 8 | Compute Manual Scorecard across 10 indicators | **Score: –10 to +10** |
| 9 | Do ML regime and Scorecard agree? Disagreement → reduce conviction | **Confirmation status** |
| 10 | Apply 5-day persistence filter — same regime for 5+ days? | **Stable regime accepted** |
| 11 | Cross-check with technicals: EMA, RSI, VIX | **Final validation** |
| **12** | **FINAL REGIME DECLARED with confidence percentage** | **Bull / Bear / Sideways + %** |

# 11. Worked Example — Full Numbers Walk-Through

The following example walks through a complete regime determination for a hypothetical trading day, using realistic feature values.

## Input: Today's Market Features

| Feature | Value | Interpretation |
|---|---|---|
| `ret_21d` | `+14.2%` | Strong 1-month gain — positive momentum |
| `ret_63d` | `+22.5%` | Excellent 3-month trend — broad rally |
| `vol_21d` | `13.1%` | Annualised volatility low — calm market |
| `vol_63d` | `16.8%` | Medium-term vol higher than recent — market calmer now |
| `vol_ratio` | `0.78` | Recent vol well below baseline (0.78 < 0.90) — bullish calm |
| `skew_21d` | `+0.42` | Positive skew — right tail dominant, no crash signal |
| `ret_z21` | `+1.85` | Momentum Z-score > 1.5 — well above 6-month average |
| `RSI-14` | `63.2` | Strong momentum, not yet overbought (below 70) |
| `MACD Hist` | `+2.8` | Clearly positive — bullish price momentum |
| `EMA20 / 60` | `23,450 / 21,200` | Golden Cross active — short trend above long trend |
| `India VIX` | `12.4` | Very low fear — institutional confidence high |
| `PCR` | `0.78` | Normal range — not extreme complacency or fear |

## Model Outputs

| Model | Output | Status |
|---|---|---|
| **K-Means (Stage 1)** | Cluster 2 — nearest to Bull centroid | **Bull** |
| **XGBoost P(Bull)** | 0.84 (84%) | **✓ Exceeds 65% threshold** |
| **XGBoost P(Sideways)** | 0.12 (12%) | **—** |
| **XGBoost P(Bear)** | 0.04 (4%) | **—** |
| **LSTM Autoencoder** | 22nd percentile — reconstruction error low | **Normal — no anomaly** |
| **Manual Scorecard** | +8 out of +10 | **8 of 10 indicators Bull** |
| **ML + Score agreement** | Both confirm Bull | **High conviction** |
| **Persistence filter** | Same Bull signal 7 consecutive days | **Stable regime** |
| **FINAL DETERMINATION** | STRONG BULL — 84% Confidence | **Full deployment** |

## Recommended Action

**STRONG BULL confirmed — 84% XGBoost confidence + Scorecard +8/10 + LSTM normal + 7-day persistence. Recommended action: Deploy 100% of EGP portfolio allocation. Favour higher-beta stocks (Beta > 1.2). Set trailing stop-loss at 5% from peak.**

*Regime signals are decision-support tools, not guarantees. Always apply appropriate position sizing. Validate this system on your own historical data before deploying real capital.*

## Quick Reference — The Complete System at a Glance

| BULL REGIME | SIDEWAYS REGIME | BEAR REGIME |
|---|---|---|
| ret_21d > +5% | \|ret_21d\| < 5% | ret_21d < –5% |
| vol_ratio < 0.90 | vol_ratio ~1.00 | vol_ratio > 1.20 |
| EMA Golden Cross | EMA crossing / flat | EMA Death Cross |
| RSI 55 – 70 | RSI 45 – 55 | RSI < 40 |
| MACD Histogram > 0 | MACD near zero | MACD Histogram < 0 |
| India VIX < 15 | VIX 15 – 25 | India VIX > 25 |
| P(Bull) ≥ 65% | max(P) < 65% | P(Bear) ≥ 65% |
| Scorecard +6 to +10 | Scorecard –5 to +5 | Scorecard –6 to –10 |
| FII net buyer | FII neutral | FII net seller |
| **ACTION: 100% deploy** | **ACTION: 65–75% deploy** | **ACTION: 30–50% deploy** |

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